

Chapter 8: Transformer Models for Hate Speech Detection in Low-Resource Languages: A Systematic Review of Methods, Datasets, and Open Challenges

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Abstract

The global rise of social media has made hate speech detection one of the most pressing challenges in natural language processing. While transformer-based models have transformed the field, their benefits have not reached languages that lack large annotated datasets, robust pre-trained models, and culturally informed tools — a gap that affects hundreds of millions of speakers worldwide. This chapter presents a systematic review of transformer-based approaches to hate speech detection in low-resource languages, drawing on studies published between 2017 and 2025. We examine the evolution of methods from classical machine learning to state-of-the-art multilingual transformers, survey available datasets and their limitations, and analyze four persistent open challenges: data scarcity and annotation quality, code-mixing and script diversity, algorithmic bias and fairness, and the emerging demand for multimodal and privacy-preserving detection systems. Key findings show that models such as mBERT, XLM-RoBERTa, MuRIL, and IndicBERT consistently outperform earlier approaches, with federated learning and cross-lingual transfer offering promising paths forward for under-resourced communities. We conclude with a research agenda to guide future work in this area.

Keywords: Transformer models, hate speech detection, low-resource languages, multilingual NLP, cross-lingual transfer, code-mixing, federated learning, explainable AI, fairness, Indian languages.

1. Introduction

Every minute, millions of posts, comments, and messages are published across social media platforms in dozens of languages. While most of this content is benign, a meaningful fraction carries hate speech — language designed to demean, threaten, or dehumanize people on the basis of who they are. Left unchecked, it amplifies social tension, contributes to real-world violence, and silences the communities it targets (Ramos et al., 2024). Automated detection systems have become indispensable tools in the effort to address this problem at scale.

The challenge, however, is far from solved — particularly outside English. The dominant assumption in NLP research has long been that data is abundant: large labeled corpora, mature pre-trained models, and well-established benchmarks. For a language like English, that assumption broadly holds. For the hundreds of other languages spoken by billions of people — including Hindi, Tamil, Bengali, Albanian, Urdu, and Bodo — it does not. These are low-resource languages: they lack the annotated training data, pre-trained infrastructure, and cultural-linguistic toolchains that high-resource languages enjoy (Pakray et al., 2025). Hate speech directed at their speakers is just as real and harmful, but automated systems can barely detect it.

Transformer models have offered the most significant advance in addressing this gap. Models such as BERT, mBERT, XLM-RoBERTa, MuRIL, and IndicBERT have demonstrated that large-scale multilingual pre-training enables effective cross-lingual transfer, allowing knowledge from resource-rich languages to be leveraged in resource-poor settings (Usman et al., 2025). Yet significant obstacles remain: training data is still scarce, code-mixing poses serious challenges to standard pipelines, cultural

specificity is poorly captured, and questions of fairness and explainability have received insufficient attention.

This chapter addresses these issues through a structured review of the literature. Our aim is to provide a clear, evidence-based account of where the field stands, what approaches have proved most effective, where the gaps lie, and what directions hold the most promise. Section 2 explains the methodology used to identify and select studies. Section 3 traces the evolution of detection methods. Section 4 surveys available datasets. Section 5 presents results and key findings. Section 6 analyzes open challenges, and Section 7 outlines future directions before a concluding section.

Before proceeding, it is worth being precise about what we mean by low-resource. A language is typically considered low-resource when it lacks one or more of the following: large annotated corpora suitable for supervised learning, robust pre-trained language models, mature NLP toolchains such as tokenizers and morphological analyzers, and an active research community producing benchmarks and shared tasks. Many of the world's languages — including most of the 22 constitutionally recognized languages of India, the majority of African languages, and numerous Balkan and Central Asian languages — meet this definition. The aggregate speaker count runs into the hundreds of millions, and the potential harm from undetected hate speech is correspondingly large (Ye, 2025; Pakray et al., 2025).

2. Review Methodology

This review follows a systematic approach to literature identification and selection. Studies were retrieved from five major academic databases: IEEE Xplore, ACM Digital Library, ScienceDirect, Google Scholar, and arXiv. Boolean query strings combined terms related to hate speech detection, transformer architectures, low-resource and multilingual NLP, and specific target language families including Indic, Balkan, Semitic, and Southeast Asian languages. The search was bounded to publications from January 2017 — when transformer-era models began emerging — through June 2025. Inclusion criteria required that a study: (1) addressed hate speech or closely related offensive content detection using machine learning or deep learning; (2) involved at least one low-resource or non-English language; and (3) reported quantitative evaluation metrics such as F1-score, accuracy, precision, or recall. Review articles, editorials, and position papers without empirical results were excluded. After applying these criteria, 48 studies were retained for close analysis, supplemented by key foundational works cited within them. Table 1 summarizes the classification framework used to organize findings.

Table 1. Classification Framework for Reviewed Studies

Dimension	Categories Covered
Methods	Lexical, Classical ML, Deep Learning, Transformer-based, Multimodal, Federated
Languages	Indic (Hindi, Tamil, Bengali, Bodo), Balkan (Albanian), Semitic (Urdu, Arabic), Others
Datasets	Binary / multiclass labels; monolingual / code-mixed; platform (Twitter, YouTube, Facebook)
Evaluation	Accuracy, Precision, Recall, F1-score, Macro-F1, AUC
Challenges	Data scarcity, class imbalance, code-mixing, bias, privacy,

Dimension	Categories Covered
	multimodality

3. Evolution of Hate Speech Detection Methods

3.1 Early Approaches: Lexical and Classical Machine Learning

The earliest automated systems for hate speech detection relied on handcrafted lexicons — curated lists of slurs, offensive terms, and flagged phrases — combined with rule-based matching and simple statistical models. While interpretable and fast to deploy, these approaches had obvious limits: they could not understand context, failed against irony and coded language, and generated high false-positive rates by flagging reclaimed or counter-speech. Supervised machine learning methods — Support Vector Machines (SVM), Logistic Regression, Naïve Bayes, and Random Forests — offered a real improvement when paired with richer feature representations: TF-IDF weighting, character n-grams, and static word embeddings such as Word2Vec, GloVe, and FastText. Cross-lingual generalization remained elusive, and performance on code-mixed or morphologically complex low-resource languages was poor.

3.2 The Deep Learning Wave

Deep learning architectures — particularly Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM) — improved performance substantially by learning hierarchical representations directly from text, without hand-crafted features. Attention mechanisms further allowed models to weight relevant parts of an input sequence dynamically. For low-resource languages, however, deep learning brought its own challenge: these architectures are data-hungry, and labeled hate speech datasets in low-resource languages are small by definition.

3.3 The Transformer Era

The introduction of BERT (Devlin et al., 2019) marked a turning point. Self-attention mechanisms and large-scale pre-training on unlabeled text created representations that could be fine-tuned effectively on modest downstream datasets — a critical advantage in low-resource settings. The multilingual extensions that followed — mBERT (trained on 104 languages), XLM-RoBERTa (100 languages, with stronger pre-training), IndicBERT (12 Indian languages), and MuRIL (17 Indian languages including code-mixed text) — enabled cross-lingual transfer: knowledge accumulated during pre-training on high-resource languages transfers to low-resource target languages during fine-tuning (Ghosh & Senapati, 2024). In practice, this means a model fine-tuned on Hindi hate speech data can provide meaningful predictions for a closely related language like Bodo, for which almost no labeled data exists. Table 2 presents a comparative summary of key transformer models reviewed in the literature.

Table 2. Key Transformer Models for Low-Resource Hate Speech Detection

Model	Languages	Key Strength	Reported Performance
mBERT	104 languages	Broad multilingual coverage; zero-shot transfer	F1: 0.72–0.89 across tasks

Model	Languages	Key Strength	Reported Performance
XLM-RoBERTa	100 languages	Stronger pre-training; better low-resource transfer	F1: ~0.86 (Albanian); +8% F1 (Urdu)
MuRIL	17 Indian languages	Trained on code-mixed Indic text	F1: 0.909 (Hindi/Marathi/Bodo)
IndicBERT	12 Indian languages	Compact; culturally grounded	Competitive on HASOC benchmarks
GPT-3.5 (few-shot)	Multiple	Strong in-context learning for low-resource	+8% F1 over XLM-R in Urdu
Dual-BERT	Hindi–English	Conversational context encoding	F1: 73.07% (code-mixed)

3.4 Context-Aware and Federated Approaches

A key limitation of standard transformer applications is that they classify posts in isolation, ignoring the conversational context in which hate speech often occurs. Dual-encoder architectures — which encode both a target post and its parent reply or thread — have demonstrated consistent improvements over single-encoder baselines. Nayak and Joshi (2021) showed that incorporating the parent tweet into a Dual-BERT encoder improved F1 to 73.07% on English–Hindi code-mixed data, compared to lower scores for concatenation-only approaches. Federated learning offers a complementary advancement, enabling models to train across multiple distributed datasets without centralizing sensitive user data. Singh and Thakur (2024) demonstrated that a federated transformer framework across 13 Indic language datasets achieved +8% accuracy and +12% F1 macro over non-federated baselines while preserving privacy — a meaningful step toward responsible deployment in under-resourced communities.

4. Datasets for Low-Resource Hate Speech Detection

High-quality labeled data is the foundation of any supervised hate speech detection system. This review identifies datasets across several language families, social media platforms, and annotation schemes. Research attention has concentrated most heavily on Indic languages — particularly Hindi, Tamil, Malayalam, Bengali, and Marathi — driven largely by the HASOC shared task series and the DravidianLangTech workshops. Beyond South Asia, a trilingual English–Spanish–Urdu dataset of 10,193 tweets (Fortuna et al., cited in Usman et al., 2025) and Albanian social media corpora (Fetahi et al., 2025) have expanded the geographic coverage of low-resource hate speech research.

It is worth pausing on what "dataset" actually means in this context. A hate speech dataset is not simply a file of social media posts with labels attached — it is the product of a series of decisions about what counts as hateful, who gets to decide, how ambiguous cases are handled, and what cultural knowledge annotators are expected to bring. These decisions profoundly shape what a model learns and, consequently, who it protects and who it harms. Studies in the reviewed literature rarely describe their annotation processes in enough detail to allow replication or critical evaluation, which is itself a problem for a field that aspires to responsible deployment (Ramos et al., 2024).

Despite this progress, structural weaknesses are widespread. Class imbalance is near-universal: hate speech typically constitutes less than 10–20% of real-world social media content, systematically suppressing recall for the class that matters most. Annotation schemes vary considerably across studies — binary labels (hate / non-hate) are simpler but lose important distinctions; multiclass schemes (e.g., hate, offensive, neutral, counter-speech) are more informative but introduce greater annotator disagreement. Perhaps most critically, datasets rarely capture the culturally specific forms of hate speech prevalent in their target communities. Caste-based discrimination, communal rhetoric, and region-specific hostility in India, for instance, are absent from most annotation guidelines (Pakray et al., 2025). Table 3 summarizes key datasets identified in the reviewed literature.

Table 3. Key Datasets for Low-Resource Hate Speech Detection

Dataset / Source	Language(s)	Size	Annotation	Key Limitation
Trilingual corpus (Fortuna et al.)	English, Spanish, Urdu	10,193 tweets	Binary	Poor Urdu coverage; class imbalance
HASOC 2019–2023	Hindi, English, Marathi	Varies (~5K–20K)	Binary / multiclass	Platform-specific; limited cultural nuance
DravidianLangTech	Tamil, Malayalam, Kannada	7K–44K	Multiclass	Significant class skew
Albanian corpus (Fetahi et al.)	Albanian	~3K comments	Binary	Small size; limited generalization
Indic multilingual (Singh & Thakur)	13 Indic languages	Distributed	Binary	Heterogeneous annotation schemes
MHSDF corpus (Prabhu&Seethalakshmi)	English + image	Multi-platform	Multiclass	No Indian language text component

5. Results and Key Findings

5.1 Transformer Models Dominate Benchmarks

Across the reviewed literature, transformer-based models consistently outperform all earlier approaches on standard evaluation metrics. XLM-RoBERTa achieves F1 scores around 0.86 on Albanian low-resource data (Fetahi et al., 2025) and an 8% F1 gain over earlier baselines on Urdu (Usman et al., 2025). MuRIL reaches a weighted F1 of 0.909 on a combined Hindi, Marathi, Bangla, and Bodo evaluation (Ghosh & Senapati, 2024). The pattern is robust: larger and more diverse pre-training corpora translate into stronger performance on fine-tuning tasks, especially in genuinely low-resource settings where target-language data is scarce.

5.2 Cross-Lingual Transfer Works Best Within Language Families

Transfer learning is most effective between typologically related languages. Hindi-to-Marathi transfer (both Indo-Aryan) and Tamil-to-Malayalam transfer (both Dravidian) consistently outperform transfers across language families. Zero-shot approaches — applying a model trained on one language directly to

another without any fine-tuning — produce meaningful but sub-optimal results. Pseudo-label fine-tuning, where a source-language model annotates unlabeled target-language data and that data is then used for fine-tuning, closes much of the gap: Zia et al. (2022, cited in Ghosh & Senapati, 2024) reported +11.1% F1 improvement over zero-shot baselines using this strategy.

5.3 Context Modeling Adds Consistent Value

Studies that incorporate conversational context — encoding the parent post, thread history, or user metadata alongside the target post — consistently report higher F1 scores than context-free alternatives. The dual-encoder BERT architecture of Nayak and Joshi (2021) improved F1 by several percentage points over single-encoder concatenation on code-mixed English–Hindi data. This finding has practical implications: real hate speech often gains or loses its hateful character depending on what preceded it, and models that ignore this context are structurally limited.

5.4 Multimodal and Federated Systems Show Strong Promise

Prabhu and Seethalakshmi (2025) demonstrated that combining CNN-based image features, LSTM text encoders, BERT embeddings, and OCR-extracted text in an attention-based fusion framework achieves ~98.5% detection accuracy on multimodal hate speech — a substantial improvement over text-only systems. Singh and Thakur (2024) showed that federated learning preserves competitive detection performance while avoiding data centralization, achieving +8% accuracy and +12% macro F1 across 13 Indic datasets. Both directions are especially relevant for low-resource settings, where data is sensitive and often community-held.

6. Open Challenges

6.1 Data Scarcity and Annotation Quality

The most fundamental bottleneck remains the shortage of large, high-quality annotated corpora for low-resource languages. Building these datasets demands annotators with native fluency, genuine cultural literacy, and training on forms of hate speech — caste discrimination, communal incitement, code-mixed slurs — that generic annotation guidelines were not designed to handle. Data augmentation via back-translation, paraphrase generation, and GAN-based synthesis can partially compensate, but risks amplifying existing biases if not carefully controlled (Pakray et al., 2025).

6.2 Code-Mixing and Script Diversity

Indian and many other low-resource online communities routinely mix two or more languages in a single post, producing varieties like Hinglish, Tanglish, and Manglish that appear in romanized script with no standardized orthography. Standard tokenizers and pre-trained embeddings are designed for monolingual input and perform poorly on such text. MuRIL was explicitly pre-trained on transliterated and code-mixed Indic data, and shows clear performance gains as a result — but most model families have not made this investment. The field also lacks standardized benchmarks for evaluating code-mixed hate speech detection specifically.

6.3 Algorithmic Bias and Fairness

Models trained on social media data inherit and can amplify the inequalities embedded in that data. They may flag content from marginalized communities at higher rates while missing hate speech directed at those same groups. Fetahi et al. (2025) and Ramos et al. (2024) both highlight bias as an underexplored

dimension in low-resource hate speech research. Explainable AI tools — particularly SHAP and LIME — are beginning to appear in detection pipelines to provide post-hoc transparency, but truly interpretable architectures remain an open research problem.

6.4 Privacy, Multimodality, and Scalability

Centralizing multilingual hate speech datasets raises serious privacy concerns, particularly when data comes from communities that are already vulnerable. Federated learning frameworks address this directly, but communication efficiency, convergence under linguistic heterogeneity, and fairness across participating nodes require further research (Singh & Thakur, 2024). On the multimodal side, memes and short-form video carry a growing share of hateful content, and text-only systems are structurally blind to these formats. Expanding coverage to images and audio while maintaining real-time inference capability at platform scale remains an open engineering and research challenge.

7. Future Research Directions

Several directions stand out as both urgent and tractable. First, the field needs systematic investment in annotated datasets for languages that currently have no hate speech corpora — including Bodo, Santali, Konkani, and Dogri — built with annotation frameworks that explicitly account for culturally specific forms of hate. These efforts should be collaborative, involving linguists, social scientists, and community representatives, and should produce standardized guidelines that enable cross-dataset comparison.

Second, context-aware architectures deserve deeper exploration. Dual-encoder and thread-level transformer models have demonstrated value, but most existing work addresses only one level of context — the immediate reply. Richer conversation modeling, incorporating longer thread histories, user behavior signals, and temporal dynamics, could yield further gains and is closely aligned with how real hate speech actually propagates online (Nayak & Joshi, 2021; Chapagain et al., 2025).

Third, federated learning combined with differential privacy guarantees represents the most principled current path toward privacy-preserving multilingual detection. Research is needed on communication efficiency, convergence properties under linguistic diversity, and fairness auditing across federated nodes. Fourth, truly interpretable models — not just post-hoc approximations — would substantially improve trust, accountability, and the ability of platform operators and policymakers to understand and contest model decisions. Finally, sustained interdisciplinary collaboration between NLP researchers, social scientists, legal scholars, platform operators, and affected communities is not optional: it is a prerequisite for systems that are technically capable, culturally grounded, and ethically sound.

8. Conclusion

This chapter has reviewed the state of transformer-based hate speech detection for low-resource languages, drawing on empirical studies from 2017 to 2025. The trajectory is clear: transformer models, particularly multilingual variants like XLM-RoBERTa, MuRIL, and IndicBERT, have moved the field far beyond what lexical and classical machine learning approaches could achieve. Cross-lingual transfer, context-aware architectures, and federated learning have each contributed meaningful advances, and multimodal detection is emerging as the next frontier.

At the same time, the review reveals that fundamental challenges persist. Annotated data is scarce and culturally thin. Code-mixed text remains difficult for standard pipelines. Algorithmic bias is real and under-studied. Privacy concerns are legitimate and need principled solutions. Progress on these fronts will

require not just better algorithms, but stronger datasets, more diverse research teams, closer engagement with affected communities, and policy frameworks that translate technical advances into accountable systems. The billions of people who communicate in low-resource languages deserve the same quality of protection from online hate that English speakers receive — and this review is intended as a contribution toward that goal.

Looking across the evidence, one conclusion stands out clearly: no single model, dataset, or method is sufficient on its own. The most effective systems combine strong pre-trained multilingual representations with task-specific fine-tuning, culturally grounded annotation, context-aware architecture design, and rigorous fairness evaluation. Building those systems is a multidisciplinary undertaking. It demands that computer scientists work alongside linguists, anthropologists, and legal scholars; that platform operators share data responsibly with researchers; and that the communities most affected by hate speech have a meaningful voice in defining what counts as harmful and how it should be handled. This chapter is offered as a foundation for that ongoing, necessary work.

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